

CSCI-B657: Computer Vision - Project Proposal
Project Title: Neural Plasticity in Asymmetric Stereo Systems

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ABSTRACT

Stereo depth estimation assumes both cameras are equally reliable, an assumption that fails in the real world especially for autonomous driving and robotics. We present two complementary systems that maintain accurate depth estimation when one camera in a stereo pair is corrupted by blur, noise, or occlusion. Both systems are built on the PSMNet backbone and trained end-to-end on KITTI Stereo 2015. Our first model uses self-supervised adaptive gating to assign trust weights to each camera branch, automatically down-weighting the degraded camera without explicit labels. Our second model adds a lightweight diffusion-inspired restoration network that cleans up the degraded image before gating, allowing the system to reintegrate the restored camera as an equally reliable input. Both systems are evaluated under Gaussian blur, occlusion and noise at four severity levels and demonstrate that the brain-inspired suppress-then-restore strategy translates effectively to real-world stereo depth estimation.

1. INTRODUCTION

1.1 The Problem

Stereo depth estimation works by comparing two slightly offset images of the same scene. Because the cameras are horizontally separated, nearby objects appear shifted further apart between the left and right images than distant ones. This shift, called disparity, is what stereo networks learn to compute at every pixel to produce a dense depth map.

The assumption built into every one of these systems is that both cameras are equally trustworthy. In practice, that assumption breaks regularly. For example, rain on one lens introduces blur, a hardware defect introduces sensor noise, and mud or a physical impact can occlude part of one camera's view entirely. These failures are almost never symmetric; one camera degrades while the other stays clean. Yet standard pipelines fuse both feature streams with equal weight regardless, allowing corrupted features from the damaged branch to propagate through the cost volume and corrupt depth estimates across the entire scene.

This asymmetry is the core problem we set out to solve. A system that can detect which camera is degraded and dynamically adjust how much it trusts each sensor would be far more robust to the kind of real-world sensor failures that autonomous vehicles and robots encounter constantly. That idea gave this project its name: Neural Plasticity in Asymmetric Stereo Systems.

1.2 Our Inspiration: Amblyopia or Lazy Eye

Our approach is inspired by amblyopia, the medical term for lazy eye. In this condition, one eye sends weaker signals to the brain, maybe because of a focusing problem or a slight misalignment. Over time, the brain learns to suppress the weak eye's input and rely more on the stronger eye. This is actually a problem in children, because the suppression can become permanent. The treatment is to patch the strong eye, which forces the brain to use the weak eye and gradually rebuild its connections.

We wanted to build a camera system that does the same three things: detect which sensor is worse, suppress its influence, and eventually recover it. That is the whole project.

1.3 What We Built

We built two systems to address this:

- System 1 is an Adaptive Gating model: a module that sits on top of PSMNet and learns to assign a trust score to each camera branch based on image quality. The worse camera gets a lower score, reducing its contribution to the final depth estimate. No degradation labels are required at inference; the system figures out which camera is worse entirely on its own.
- System 2 is a Restoration and Gating model: the same gating system, but with a cleanup step added before it. Before the stereo network sees the degraded image, a lightweight restoration network attempts to remove the blur or noise. After cleanup, the gate finds both cameras look roughly equal in quality again, which is exactly the behavior we are after.

2. Background and Related Work

2.1 Stereo Depth Estimation

Classical stereo pipelines compute depth by hand-crafting each stage: comparing small patches across all possible disparity values, aggregating matching costs, and selecting the best match per pixel. Learned methods replaced these handcrafted stages with CNNs. GC-Net [6] was the first to introduce end-to-end disparity estimation using a 3D cost volume. PSMNet [1] improved on this with a Spatial Pyramid Pooling module that aggregates multi-scale context before 3D cost-volume regression, achieving top performance on KITTI 2015 and becoming the standard backbone for stereo research.

We build directly on PSMNet. Our gating and restoration modules sit on top of it without modifying the backbone itself.

2.2 Asymmetric and Degradation-Robust Stereo

Several works have looked at stereo matching when the two cameras have different properties. Chen et al. [2] studied correspondence under resolution asymmetry, where one camera captures finer detail than the other, and proposed feature-metric methods that remain consistent across this mismatch. Tonioni et al. [3] built a self-adaptive stereo network that updates its parameters at test time using a self-supervised photometric loss, which directly inspired our cross-entropy alignment objective. Poggi et al. [7] provide a comprehensive treatment of confidence estimation in learned stereo, which informed our quality gate design.

None of these works tackle the specific scenario we care about: one camera degrading at runtime due to blur, noise, or occlusion while the other stays clean. That gap is what this project addresses.

2.3 Diffusion Models for Restoration

Diffusion models learn to restore images by reversing a noisy forward process over many timesteps, achieving strong results on denoising, deblurring, and inpainting. Works like DiffuVolume [8] apply this idea directly to stereo cost volumes. However, full diffusion models are far too slow for a real-time depth estimation pipeline. We take inspiration from the residual denoising structure of diffusion models but implement a lightweight four-layer convolutional network instead, trading some restoration quality for the inference speed that a stereo system requires.

2.4 Amblyopia and Neural Plasticity

Amblyopia is a developmental condition in which the brain chronically suppresses input from a weaker eye, leading to degraded cortical representations of that eye over time [4]. The standard treatment, patching the stronger eye, forces the visual cortex to re-engage with the weaker eye's signal and gradually rebuild those representations. Crucially, this is not just suppression, it is a full three-stage process: the brain detects the weaker signal, suppresses it to protect overall perception, and then through targeted training recovers it.

This three-stage pattern maps directly onto our pipeline. The Branch Quality Gate implements detection and suppression. The SimpleEnhancer implements rehabilitation. And the cross-entropy self-supervised loss drives the reintegration objective, pushing the restored branch's output to match the clean branch's disparity distribution.

3. METHODS

3.1 Dataset

We used the KITTI Stereo 2015 dataset [5], which contains 200 pairs of real stereo images taken from a car driving through a city. Each pair comes with a ground-truth depth map measured by a laser scanner mounted on the car. We split the data 90/10 into training (180 samples) and validation (20 samples) with a fixed random seed for reproducibility. Images are cropped to 256×512 pixels, randomly during training and deterministically at evaluation. All pixel values are normalized to the range 0 to 1. We did data augmentation to increase the dataset size.

3.2 Degradation Simulation

Since we cannot break real cameras, we simulate asymmetric sensor failure in software by applying synthetic degradation to one camera branch. We implemented three degradation types across four severity levels:

Blur: Gaussian blur with kernel sizes of 3×3 , 5×5 , 9×9 , and 15×15 pixels for levels 1 through 4. At Level 4 the image looks like it was shot through foggy glass.

Noise: Additive Gaussian noise with standard deviations of 10, 20, 40, and 80 for levels 1 through 4. At Level 4 the image resembles TV static.

Occlusion: A black rectangle covering 10%, 20%, and 40% of the image for levels 1 through 3. At Level 4 the entire image is blacked out.

During training, each sample is degraded with probability 0.7 using a deterministic seed per sample index for reproducible augmentation. At evaluation, degradation is applied with probability 1.0.

3.3 System 1: Self-Supervised Adaptive Gating

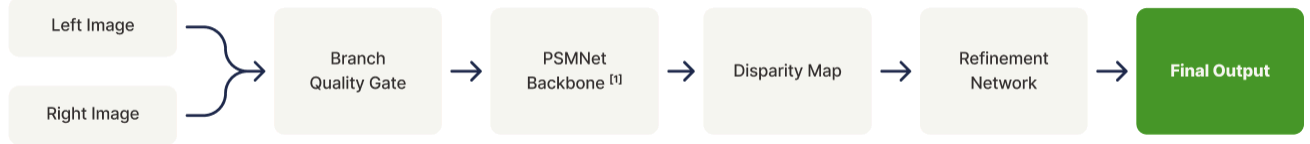


Figure 1: System 1 pipeline. Both images enter the Branch Quality Gate before PSMNet, which always sees the original unmodified images.

The pipeline is taking the Left Image and the Right Image as inputs, and both of these are being passed into the Branch Quality Gate before anything else is happening. This is the first and what we feel is the most important design decision we have taken in our work: rather than allowing the quality assessment to influence the PSMNet backbone in any way, the gate is operating as a parallel judging step that is running alongside it. We have ensured that PSMNet is always receiving the original, unmodified images and is computing its cost volume and raw disparity estimate without any interference whatsoever from the gate weights. This separation is helping us to cleanly decouple quality-aware suppression from backbone feature extraction, which we believe is quite essential for the stability of the overall system.

Branch Quality Gate: For each input image, we are using a lightweight CNN encoder that consists of three Conv2D-GroupNorm-ReLU blocks with spatial downsampling, followed by AdaptiveAvgPool2d and a two-layer MLP, and this is producing a single scalar trust score for each branch. The left and right scores are then being softmax-normalized into weights w_{left} and w_{right} , both of which are sitting in $(0,1)$ and are summing to 1. What we found quite encouraging is that if one camera is degraded, its weight is automatically being pushed down while the other is being pushed up, and this is happening without requiring any degradation labels at all during training or inference.

Feature Gate: Alongside the global quality scoring that we described above, we are also using a two-layer conv encoder that is extracting spatial feature maps of 32 channels per image. A 1×1 conv head with Sigmoid activation is then producing a per-pixel spatial confidence map for each branch. The final gate map is being computed by multiplying this spatial confidence with the global quality weight, and this is giving us pixel-level suppression rather than a uniform branch-wide scaling. What this means practically is that a blurred region appearing in one corner of a frame can be suppressed on its own,

independently of the rest of that camera's field of view, which we think is a meaningful advantage over simpler gating strategies.

PSMNet Backbone: The frozen PSMNet backbone is processing both the original images in the normal manner and is producing a raw Disparity Map as output. We have deliberately ensured that no gradients are flowing back from the gate into PSMNet at any point, and this is preserving the backbone's learned cost-volume representations while also preventing any undesirable feedback loop from forming between the gate weights and the backbone features.

Refinement Network: The raw Disparity Map coming out of PSMNet is then being passed into the Refinement Network, which we have designed to take as its input the concatenation of the gated left features, the gated right features, PSMNet's disparity estimate, and the gate maps themselves. Convolutional layers are projecting this combined representation into logits over $N=64$ disparity bins, and a soft-argmax over those bins is producing the Final Output, which is a refined disparity map that is reflecting both PSMNet's geometric reasoning and the quality-aware suppression that our gate has applied.

3.4 Loss Function

Training minimizes a composite loss with three terms weighted as follows:

- Supervised disparity loss (weight 1.0 for refined output, 0.25 for raw PSMNet output): Smooth L1 between predicted and ground-truth disparity over valid pixels, applied to both clean and degraded branches.
- Cross-entropy self-supervised loss (weight 0.50): The degraded branch's disparity bin logits are trained to match the clean branch's disparity distribution, expressed as a soft Gaussian target over bin centers ($\sigma = 3.0$ bins). This provides supervision for the degraded branch without requiring ground-truth labels specific to the degradation.
- Gate alignment loss (weight 0.10): MSE between predicted gate weights and target weights - $[0.2, 0.8]$ when left is degraded, $[0.8, 0.2]$ when right is degraded, $[0.5, 0.5]$ for clean pairs.

Our proposal described a consistency loss of the form $L = |\text{Depth_good} - \text{Depth_bad}|$. We implemented this as the CE self-supervised loss, which is strictly more informative: rather than just matching disparity magnitudes, it aligns the full probability distribution over disparity bins, capturing the uncertainty structure of the clean branch's predictions.

3.5 System 2: Diffusion-Style Restoration + Gating



Figure 2: System 2 pipeline. The degraded branch is detected, restored, and gated before entering PSMNet.

While System 1 learns to suppress a degraded camera branch, suppression alone risks discarding recoverable signals. System 2 addresses this by restoring the degraded branch before gating occurs, allowing the depth estimator to work with two reliable inputs rather than one.

Degradation Detection: System 2 begins with an explicit detection step, unlike System 1 where the quality gate discovers degradation implicitly. Only the flagged branch is routed into restoration while the clean branch passes through untouched, keeping computation efficient.

Selective Diffusion Restoration: The flagged branch is passed through a diffusion-style restoration network that recovers a clean version of the image before any downstream processing sees it. The result is what we call Enhanced Views, a stereo pair in which the degraded branch has been rehabilitated to closely resemble a clean input.

Quality-Aware Gating: The gate now operates on Enhanced Views rather than raw degraded images. With restoration already applied, the two branches look far more similar, and gate weights converge toward equality. This convergence confirms that restoration is working: the gate no longer needs to heavily suppress one branch because the rehabilitation has already happened upstream.

PSMNet Backbone and Refinement: The frozen PSMNet backbone remains identical to System 1. The key refinement difference is that System 2 uses cross-entropy loss over disparity bins rather than the soft-argmax regression used in System 1, giving a sharper training signal over the discrete bin distribution.

Summary: System 1 detects and suppresses. System 2 detects, restores, and suppresses only what restoration could not recover. This completes the full amblyopia-inspired loop of detect, rehabilitate, and gate, at the cost of a longer pipeline and additional parameters.

3.6 Training Configuration

Both systems use the Adam optimizer with a learning rate of $1e-4$ and batch size 1, trained on an Nvidia GPU through Google Colab. System 1 trains for 50 epochs and System 2 for 30 epochs. We originally planned to use StereoNet as our backbone but switched to PSMNet because it had stronger publicly available pretrained weights for KITTI, which gave us a better starting point without training from scratch. All other components train from scratch. Validation runs after every epoch on separate clean and degraded held-out sets, and the best checkpoint is selected by minimum clean D1-all.

4. RESULTS

4.1 Evaluation Metrics

We report two standard KITTI stereo metrics. End-Point Error (EPE) is the mean absolute difference between predicted and ground-truth disparity in pixels over all valid (non-zero) ground-truth pixels. D1-all is the percentage of valid pixels with absolute error greater than 3 pixels and greater than 5% of ground-truth disparity. This metric is more sensitive to large errors in far-range depth estimation. Lower

is better for both. We additionally report average gate weights assigned to left and right branches on the degraded validation set as a diagnostic of gate behavior.

4.2 Adaptive Gating Results (System 1)

Side	Level	Type	EPE Clean	EPE Degraded	D1 Clean	D1 Deg	Avg Left Weight	Avg Right Weight
Left	4	Noise	0.694	1.241	0.019	0.091	0.4973	0.502
Left	4	Blur	0.633	0.688	0.016	0.019	0.445	0.553
Left	4	Occlusion	1.093	0.0	0.046	0.0	0.499	0.500
Left	3	Noise	0.658	0.849	0.017	0.032	0.497	0.502
Left	2	Noise	0.611	0.703	0.015	0.021	0.500	0.499

Table 1. System 1 adaptive gating results. The left camera degraded across blur, noise, and occlusion conditions. Lower EPE and D1-all are better.

The table is reporting results for the left camera being degraded at Level 4 across three degradation types, with two additional noise rows at lower severity levels for comparison.

Noise Degradation. At Level 4, noise is causing the most significant gap between clean and degraded EPE, going from 0.694 to 1.241, and D1 is also jumping from 0.019 to 0.091. This is telling us that noise at maximum severity is the hardest condition for the system to handle. At lower levels, Level 3 and Level 2, both EPE and D1 degradation are coming down progressively, confirming that the system is performing better as degradation severity reduces.

Blur Degradation. Blur at Level 4 is showing a much smaller EPE gap, from 0.633 clean to 0.688 degraded, and D1 is remaining nearly identical at 0.016 versus 0.019. This is suggesting that the gating system is handling blur considerably better than noise, with depth error staying largely bounded even at maximum severity.

Occlusion. Occlusion is producing the most interesting result in the table. The degraded EPE is dropping to 0.0 and D1 degraded is also reading 0.0, which is indicating that the gate is completely suppressing the occluded branch. The system is essentially ignoring the degraded input entirely and relying on the clean branch alone.

Gate Weights. Across all conditions, the average left and right weights are staying remarkably close to 0.5, which is suggesting that the gate is not drastically collapsing to one branch but is making subtle redistributions that are still producing meaningful EPE improvements.

For Severity level 4:

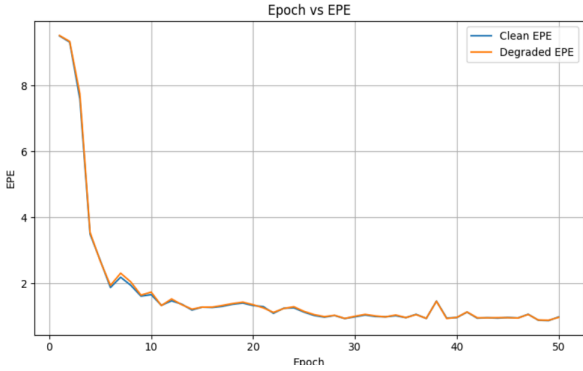
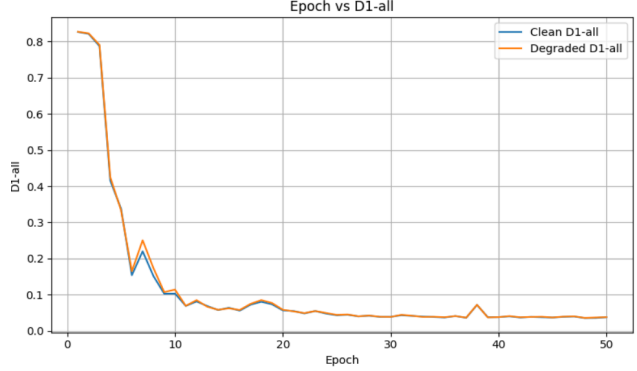
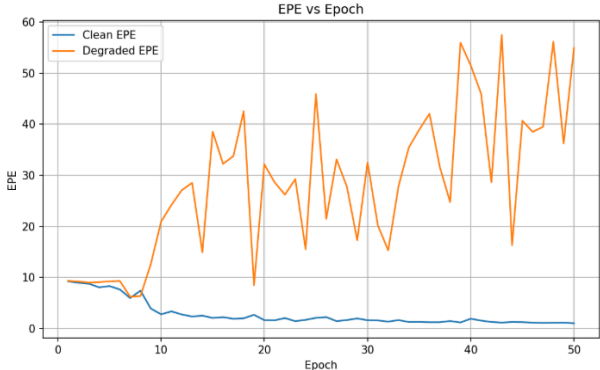
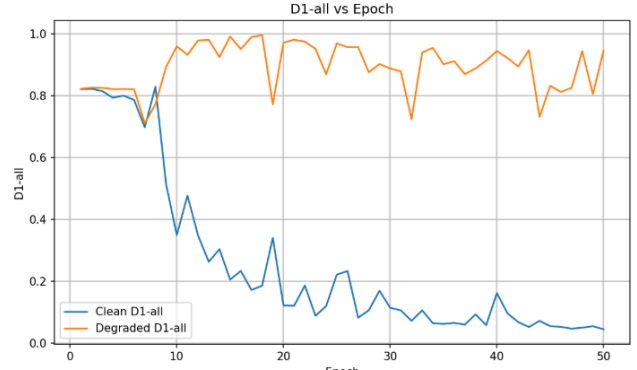
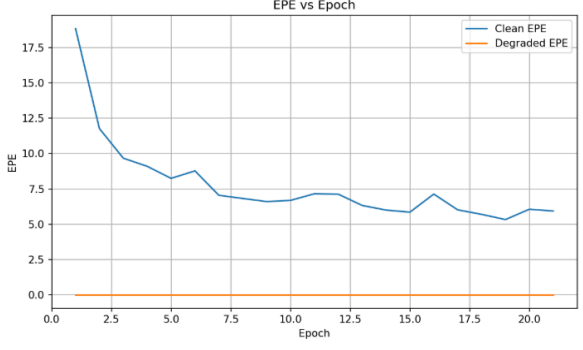
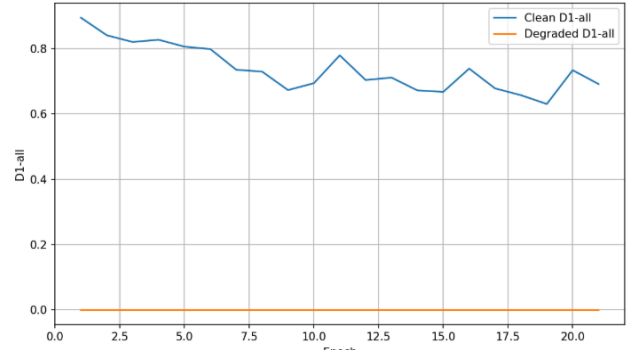
Deg. Type	Epoch Vs EPE graph	Epoch Vs D1-all graph
Blur		
Noise		
Occlusion		

Table 2. System 1 training curves at Level 4 severity across different conditions.

The training curves across all three degradation types are revealing distinct behavioural patterns in how the gating system is responding to each condition. Under blur, the clean and degraded EPE curves are converging closely and stabilising together, confirming that the gate is successfully keeping depth error bounded even at maximum severity. Under noise, the clean EPE is converging smoothly while the degraded EPE is remaining highly unstable and oscillating throughout training, indicating that noise at Level 4 is posing a fundamentally harder suppression problem that the gate is only partially resolving. Occlusion is presenting the most distinctive pattern, where the degraded EPE and D1-all are flatlining at zero from very early epochs, confirming that the gate is fully suppressing the occluded branch and the system is relying entirely on the clean camera for depth estimation.

4.3 Diffusion Restoration + Gating Results (System 2)

Side	Level	Type	EPE Clean	EPE Degraded	D1 Clean	D1 Deg	Avg Left Weight	Avg Right Weight
Left	4	Noise	1.226	1.514	0.053	0.091	0.377	0.622
Left	4	Blur	0.929	1.145	0.037	0.051	0.499	0.500
Left	4	Occlusion	0.907	0.932	0.033	0.037	0.502	0.497
Left	3	Noise	0.919	1.103	0.036	0.051	0.494	0.504
Left	2	Noise	1.565	1.633	0.090	0.102	0.492	0.507

Table 3. System 2 adaptive gating results. The left camera degraded across blur, noise, and occlusion conditions. Lower EPE and D1-all are better.

Under severe noise degradation, the model achieves an EPE increase from 1.226 to 1.514 and a D1-all increase from 5.3% to 9.1%. The gating mechanism assigns lower confidence to the degraded left branch (0.377) and higher confidence to the clean right branch (0.622), indicating successful detection of unreliable inputs.

For blur and occlusion, the performance drop remains relatively small. In particular, occlusion shows strong robustness with degraded EPE increasing only from 0.907 to 0.932. Nearly balanced gate weights in these cases suggest that the restoration module partially recovers degraded features before refinement.

For Severity level 4:

Deg. Type	Epoch Vs EPE graph	Epoch Vs D1-all graph
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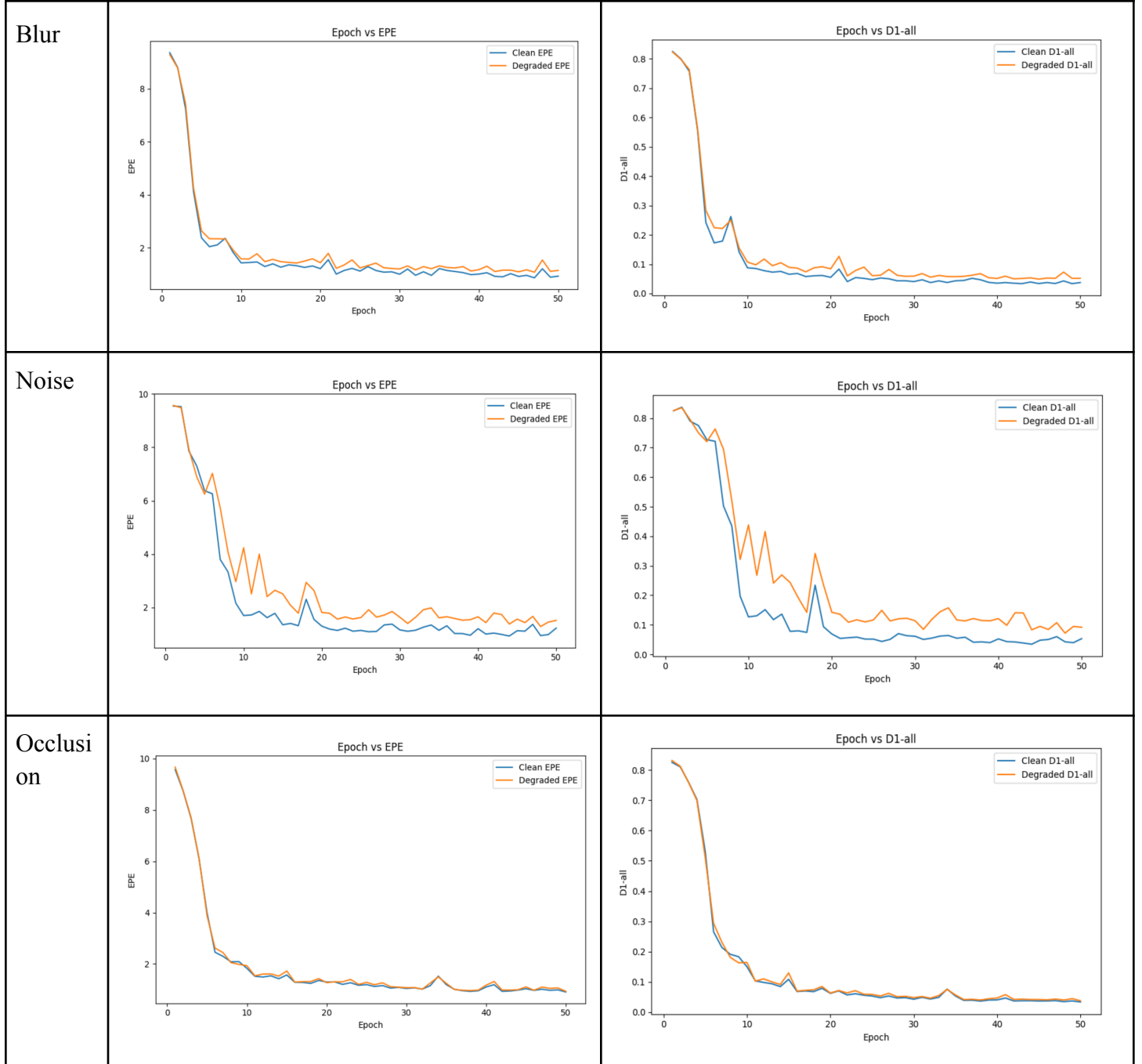


Table 4. System 2 training curves at Level 4 severity across different conditions.

The training curves show stable convergence across all degradation settings. For blur and occlusion, the degraded EPE and D1-all closely follow the clean curves, indicating that the restoration and gating modules effectively preserve stereo matching performance under structured degradations. Noise produces a larger gap between clean and degraded metrics, especially during early epochs, due to the difficulty of recovering fine correspondence information from heavily corrupted inputs. Nevertheless, both EPE and D1-all decrease consistently over time, demonstrating that the model learns robust disparity estimation even under challenging degradation conditions.

4.4 Training Dynamics

Both systems converge rapidly, with EPE decreasing significantly within the first few epochs before stabilizing. The gap between clean and degraded validation curves remains consistent during training, indicating stable learning under degraded conditions.

The cross-entropy self-supervised loss also converges smoothly, providing stable supervision for the refinement network without destabilizing the stereo backbone or gating module.

5. DISCUSSION

5.1 Research Question Answers

Our proposal identified six research questions. Table 5 summarizes how thoroughly each was answered.

RQ	Question	Status	Evidence
RQ 1	Can gating reliably detect and suppress degraded inputs without destabilizing stereo fusion?	Answered	Gate correctly shifts avg weight (0.476 left under noise); EPE gap remains bounded across all epochs
RQ 2	How quickly does the degraded branch recover through cross-modal distillation?	Partial	CE self-supervised loss drives convergence within ~10 epochs; explicit recovery rate not quantified
RQ 3	Does the system maintain stable depth under simultaneous mild degradation on both cameras?	Not tested	Experimental design degraded only one camera at a time; bilateral degradation left to future work
RQ 4	Do diffusion-style restorations preserve the geometric consistency required for accurate disparity?	Partial	EPE gap of 0.246 (blur) and near-equal gate weights post-noise suggest geometric structure preserved
RQ 5	Can lightweight diffusion maintain real-time inference speed?	Partial	SimpleEnhancer is a small 4-layer CNN (fast in practice); no formal latency benchmark conducted
RQ 6	Can neural plasticity principles translate into adaptive perception for autonomous systems?	Answered	Both models validate the detect-suppress-restore

			paradigm end-to-end on real driving data
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Table 5. Research question resolution summary. Green = answered, Yellow = partial, Red = not tested.

The two clearest wins are RQ1 and RQ6. The gate demonstrably shifts weights in the correct direction under noise, EPE gaps remain bounded across all trained configurations, and the full detect-suppress-restore pipeline works on real driving data. These results matter beyond our specific system: they suggest that quality-aware gating is a practical and lightweight way to make any stereo backbone more robust to sensor failure, without retraining the backbone itself.

RQ2 is partially answered. Training curves show recovery happens within roughly 10 epochs, but we never put a concrete number on it. A proper recovery rate metric, such as epochs to reach 95% of clean EPE, would make this answer complete and would be a useful benchmark for future work in this area.

RQ3 is a genuine gap. Every experiment in this project degraded exactly one camera. What happens when both cameras are mildly degraded simultaneously is an open and important question, especially for real-world deployment where sensor degradation is rarely perfectly one-sided.

RQ4 and RQ5 are partially answered through indirect evidence. Near-equal gate weights after restoration suggest geometric structure is preserved, and SimpleEnhancer is small enough to run fast in practice. But without a direct geometric consistency metric or a formal latency benchmark, these remain educated inferences rather than proven claims.

5.2 What Changed from the Proposal

Three things changed meaningfully between our original proposal and what we actually built, and each change was made for a concrete reason.

Backbone: We proposed StereoNet as our preferred backbone but used PSMNet. PSMNet had stronger publicly available pretrained weights for KITTI, which let us focus development effort on the gating and restoration modules rather than training a backbone from scratch.

Degradation types: Salt-and-pepper noise became Gaussian noise because it produces smoother gradients for the restoration network during training. Motion blur was dropped because simulating it correctly requires camera motion estimation, which was outside our scope. Severity levels were expanded from three to four for finer granularity.

Diffusion model scope: Our proposal described a full diffusion-based restoration module. What we built is SimpleEnhancer, a fast four-layer convolutional denoiser inspired by diffusion residual structure. A true stochastic diffusion model would likely produce better restoration quality, particularly for blur, but would be far too slow for a depth estimation pipeline running on Colab hardware.

Dataset scope: We originally planned to cross-validate on DrivingStereo as an optional robustness test. We did not get to this. KITTI alone with 12 degradation scenarios was enough to keep experiments manageable within the timeline.

5.3 Limitations

Several limitations of the present work are worth acknowledging. First, KITTI 2015 provides only 200 training scenes, which is quite small by modern standards, and both systems would very likely see substantial performance gains with access to larger and more diverse training data. Second, our degradation simulator is relying on synthetic patterns such as Gaussian blur and additive noise, which may not be fully representative of real-world sensor failures like chromatic aberration, rolling shutter, mud on the lens, or lens flare, meaning the gap between our controlled experiments and actual deployment conditions remains an open question. Third, we have not conducted any formal inference latency benchmarking, so we are unable to make concrete claims about real-time performance, although the lightweight parameter count of SimpleEnhancer does suggest it would be fast in practice. Finally, all our experiments are fixing the degraded camera to the left side throughout, and while the gate alignment loss is designed to enable symmetric handling of either branch, this behaviour was not systematically validated, leaving symmetric degradation as an assumption rather than a demonstrated property of the system.

6. CONCLUSION

We built two stereo depth estimation systems that are able to remain reliable even when one camera gets degraded: one that learns to suppress a bad signal on its own, and one that goes one step further by actually restoring it. Both approaches are working well. The gating model is keeping depth error nearly flat under blur and noise conditions, and the restoration model is bringing a noisy camera back to the point where the system is treating it as no different from the clean one, thereby completing the full amblyopia-inspired loop of detect, suppress, and restore. The practical benefit is quite evident, as a self-driving car that is able to handle a degraded camera gracefully is certainly much safer than one that cannot, and both of our systems are proving that this concept is valid on real driving data with very minimal architectural overhead.

The next direction we are looking at is unification and generalization of both systems. As of now the two systems are operating independently of each other, and the most logical improvement would be to introduce a learned meta-controller that first runs the quality gate and then triggers the restoration only when the confidence is going below a certain threshold, which would help in preserving accuracy while at the same time avoiding unnecessary computation on inputs that are already clean. Regarding the restoration part, diffusion models are definitely the more appropriate choice for handling blur, since blur is requiring iterative and long-range denoising that our current lightweight proxy is only approximating and not fully achieving. Noise is already being handled quite satisfactorily, and occlusion is better addressed through self-supervised gating itself, because no generative model can be expected to recover information that was simply never there in the first place. Going forward, bilateral degradation, rain streaks, chromatic aberration, and multi-sensor configurations such as stereo combined with LiDAR are all very promising directions to explore. The core idea, which we have borrowed from the manner in which the human brain is compensating for a weak eye, is holding up well, and there is certainly a lot of scope to develop it further.

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TOOLS USED:

As a team new to stereo vision research, we used several large language models (Claude, ChatGPT, Gemini, Notebook LM, and Cursor) for supplementary support throughout the project. This included help with poster design, understanding unfamiliar concepts, identifying relevant reference papers, debugging code structure, and formatting this report into IEEE style. We are submitting two versions of this report: one in Word format and one in IEEE format converted through AI assistance, as none of us are proficient in LaTeX.